NEIL M. DWIVEDI

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Versatile back-end software engineer with extensive experience architecting, development and modernizing enterprise-scale applications. Experience writing Python scripts, developing microservices with Spring and Java and web development using AngularJS in Agile environments. Leveraged AI to identify and implement process efficiencies and automation opportunities. Strong communication, presentation and collaboration skills.

CLEARANCES

- DOD – Secret, DHS – EOD Granted IT High (6C), IRS – FINAL Moderate (MBI)

TECHNICAL SKILLS

Languages: Java, Python, NodeJS, AngularJS, HTML, CSS, XML, JavaScript, SQL

Technologies: Agile, Spring Boot & Batch AWS, Git, Linux, pgAdmin, DBeaver, Postgres, MongoDB

WORK EXPERIENCE

The MITRE Corporation (*May 2020 – June 2025)*

Intermediate Software Engineer

Common Weakness Enumeration Website Usability

- Acted as product owner and maintainer for cwe.submission.mitre.org and associated AWS instances
- Reduced deployment time by 80% by developing Python scripts to optimize deployment pipeline

IRS Customer Account Data Engine 2

- Developed proof-of-concept microservices to modernize tax processing architecture using Java & Spring Batch
- Built and maintained optimized tax processing logic proof-of-concept to modernize legacy architecture using Java and Spring Batch

Enterprise Architecture Baseline – Simplified (EnABLS)

- Architected and built new visualization tools for EnABLS using AngularJS improving process visibility for all sponsors
- Deployed regular updates for new EnABLS versions to 7 unique sponsor environments

IRS Independent Verification & Validation

- Audited code quality and development protocols for multiple IRS projects resulting in identification of enterprise-wide issues which were acted upon by IRS leadership and development teams
- Led code quality audit for 2 projects resulting in 3-5 technical process improvements per project

Easy Room Reservation App

- Developed ReactJS app to streamline employee interactions with TRIRIGA room reservation system to improve accessibility and efficiency during return-to-office

Sandalwood Tool Suite

- Developed new web-scraping tool using Python and JavaScript allowing users to select specific fields to be scraped from any web source

Structured Threat Information Expression (STIX) for AIS FIRR

- Designed and implemented new STIX extensions using Python

Transform with Resilience during Upgrades to Socio-Technical Systems (TRUSTS)

- Lead development of TRUSTS informational website

- Co-lead and published 3 studies on TRUSTS framework for system resilience

Department Hiring

- Lead interviews and phone screened internship candidates
- Modernized interview processes to adapt to current software development practices

Confidential Department of Defense Project

- Led dashboard development to visualize DevSecOps pipeline for ARCUS cloud environment

EDUCATION

B.S. in Computer Science, Texas A&M University, College Station, TX (*August 2021)*
Microsoft AI & ML Engineering certificate – in progress

Publications

Augmenting Agile Software Development to Improve Resilience during Real-World Operation
32nd Annual Symposium of the International Council on Systems Engineering (INCOSE), Detroit, MI
Emily Barrett; Steve Dorton; **Neil M Dwivedi**; Dr. Kelly Neville; Dr. Kris Rosfjord
Professional Society: International Council on Systems Engineering (INCOSE)

System Resilience: A framework and tools for new technology integration
Human Factors and Ergonomics Society Healthcare Symposium, New Orleans, LA
Dr. Kelly Neville; Dr. Poornima Madhavan; Dr. Bianica S Pires; Booth, Matthew; Emily Barrett; Heather E Rosso; **Neil M Dwivedi**; Dr. Kris Rosfjord; Patterson, Emily

A system resilience framework for improving the design of advanced technology
Department of Defense: Measuring Human-Machine (AI) Interactions TEM
Dr. Kelly Neville; Dr. Bianica S Pires; Dr. Poornima Madhavan; Booth, Matthew; Heather E Rosso; **Neil M Dwivedi**; Dr. Kris Rosfjord; Patterson, Emily